

1.

2.

3.

4.

5.

6.
-

1.

1.1

1.2

" "

1.3

-
- App

1.4

	/	

2.

2.1

2.1.1

flowchart LR

```
A[用户注册/登录] --> B[创建学习任务]
B --> C[执行学习任务]
C --> D[完成打卡]
D --> E[查看数据分析]
B -. -> F[系统推荐任务]
C -. -> G[任务提醒推送]
```

2.1.2

flowchart TD

```
A([开始]) --> B{选择创建方式}
B -->|自定义| C["📝 填写任务表单<br/>主题/描述/截止时间<br/>优先级/复习选项"]
B -->|推荐模板| D["📁 浏览推荐模板<br/>选择并修改"]
C --> E["💾 保存任务"]
D --> E
E --> F["📋 任务出现在任务列表"]
F --> G["📖 用户执行学习任务"]
G --> H{是否完成?}
H -->|否| I["🕒 等待/接收提醒"]
I --> G
H -->|是| J["✅ 点击打卡按钮"]
J --> K["📝 打卡记录保存<br/>任务状态更新为已完成"]
K --> L["📊 查看数据分析/日历"]
L --> M([结束])
```

```
style A fill:#4CAF50,color:#fff
style M fill:#4CAF50,color:#fff
style E fill:#2196F3,color:#fff
style J fill:#FF9800,color:#fff
style K fill:#FF9800,color:#fff
```

2.1.3

flowchart TD

```
A["🕒 系统定时任务<br/>定时扫描未完成任务"] --> B{"临近截止时间?"}
B -->|已逾期| C["🔴 紧急提醒<br/>立即推送"]
B -->|今天到期| D["🟡 到期提醒<br/>当天推送"]
B -->|3天内到期| E["🟢 温和提醒<br/>提前通知"]
B -->|还早| F["无需提醒<br/>等待下次扫描"]
C --> G["📧 生成提醒记录<br/>推送通知给用户"]
D --> G
E --> G
```

F --> A
G --> H["🔔 用户查看提醒"]
H --> I["📖 前往完成任务"]
I --> A

style C fill:#f44336,color:#fff
style D fill:#FF9800,color:#fff
style E fill:#4CAF50,color:#fff
style G fill:#2196F3,color:#fff

2.2 User Story

ID		
US-01		
US-02		
US-03		

ID		
US-04		
US-05		
US-06		
US-07		
US-08		

ID		
US-09		" "
US-10		

US-11		
-------	--	--

ID		
US-12		
US-13		/
US-14		

ID		
US-15		
US-16		

3.

3.1 Use Case Diagram

```
flowchart TD
    subgraph System["📦 学习养成计划系统"]
        UC01[注册账号]
        UC02[登录系统]
        UC03[编辑个人资料]
        UC04["📝 创建学习任务<br/>(含编辑/删除/筛选)"]
        UC05["📅 从推荐模板添加任务"]
        UC06["✅ 完成任务打卡"]
        UC07["📅 查看打卡记录"]
        UC08["📊 查看数据分析<br/>(进度条/图表/日历)"]
        UC09["🔔 接收任务提醒"]
    end

    Actor((👤 学习者))
    Scheduler[🕒 定时任务]

    Actor --> UC01
    Actor --> UC02
    Actor --> UC03
    Actor --> UC04
    Actor --> UC06
```

```

Actor --> UC07
Actor --> UC08
UC04 -.->| 《extend》 | UC05
UC06 -.->| 《include》 | UC07
Scheduler --> UC09
UC09 -.-> Actor

style System fill:#e3f2fd,stroke:#1565c0
style Actor fill:#fff3e0,stroke:#e65100
style Scheduler fill:#f3e5f5,stroke:#7b1fa2

```


	UC-01		
	UC-02		
	UC-03		
	UC-04		
	UC-05		UC-04
	UC-06		
	UC-07		
	UC-08		
	UC-09		

3.2 DFD — Data Flow Diagram

3.2.1 Context Diagram

```

flowchart LR
    User["👤 学习者<br/>(源/宿)"]
    subgraph System["📦 学习养成计划系统 (0)"]
        Main["学习养成<br/>计划系统"]
    end
    User -->|"任务操作请求<br/>打卡操作请求<br/>查询统计请求<br/>登录/注册请求"| Main
    Main -->|"提醒通知<br/>打卡确认<br/>统计数据<br/>登录结果<br/>任务列表/详情"| User

```

```
style System fill:#e3f2fd,stroke:#1565c0
style User fill:#fff3e0,stroke:#e65100
```

3.2.2 0

```
flowchart TD
    User[("👤 学习者<br/>(源/宿)")]

    subgraph System["🏠 学习养成计划系统 - 0层"]
        P1["1. 用户认证管理"]
        P2["2. 任务管理"]
        P3["3. 打卡管理"]
        P4["4. 数据分析与提醒"]
        P5["5. 推荐模板管理"]
    end

    subgraph Storage["💾 数据存储"]
        D1["D1<br/>用户表"]
        D2["D2<br/>任务表"]
        D3["D3<br/>打卡记录表"]
        D4["D4<br/>提醒记录表"]
        D5["D5<br/>推荐模板表"]
    end

    User --> P1
    User --> P2
    User --> P3
    User --> P4
    User --> P5
    P1 -->|"用户信息"| D1
    P2 -->|"任务数据"| D2
    P3 -->|"打卡数据"| D3
    P4 -->|"提醒数据"| D4
    P5 -->|"模板数据"| D5
    P1 --> User
    P2 --> User
    P3 --> User
    P4 --> User
    P5 --> User

    style System fill:#e3f2fd,stroke:#1565c0
    style Storage fill:#f3e5f5,stroke:#7b1fa2
    style User fill:#fff3e0,stroke:#e65100
```

1		

2		
3		
4		
5		

3.3 Sequence Diagram

3.3.1

```
sequenceDiagram
    actor User as 👤 学习者
    participant FE as 🖥️ 前端Vue
    participant BE as ⚙️ 后端API
    participant DB as 🗄️ 数据库
    participant Scheduler as 🕒 定时任务

    Note over User,DB: 1. 用户登录
    User->>FE: 输入用户名密码
    FE->>BE: POST /api/user/login
    BE->>DB: 查询用户信息
    DB-->>BE: 返回用户数据
    BE--->>FE: 返回 JWT Token
    FE--->>User: 进入主页

    Note over User,DB: 2. 创建任务
    User->>FE: 填写任务表单并提交
    FE->>BE: POST /api/tasks
    BE->>DB: INSERT 任务数据
    DB-->>BE: 插入成功
    BE--->>FE: 返回任务详情
    FE--->>User: 任务出现在列表

    Note over User: 3. 执行学习...

    Note over User,DB: 4. 完成打卡
    User->>FE: 点击打卡按钮
    FE->>BE: POST /api/checkin/{taskId}
    BE->>DB: INSERT 打卡记录
    BE->>DB: UPDATE 任务状态 → DONE
    DB-->>BE: 更新成功
    BE--->>FE: 返回打卡成功
    FE--->>User: 显示打卡成功提示

    Note over Scheduler,DB: 5. 定时提醒
    Scheduler->>DB: 扫描未完成任务
    DB-->>Scheduler: 返回临近截止的任务
    Scheduler->>DB: INSERT 提醒记录
```

3.3.2

```
sequenceDiagram
    actor User as 👤 学习者
    participant FE as 🖥️ 前端Vue
    participant BE as ⚙️ 后端API
    participant DB as 🗄️ 数据库

    Note over User,DB: 进入数据分析页
    User->>FE: 点击"数据分析"
    FE->>BE: GET /api/statistics/overview
    BE->>DB: 查询任务统计(COUNT/GROUP BY)
    DB-->>BE: 返回统计结果
    BE->>DB: 查询打卡统计
    DB-->>BE: 返回打卡统计
    BE-->>FE: 返回 JSON {完成率, 任务数, 打卡数...}
    FE-->>User: 渲染进度条 + 统计表

    Note over User,DB: 查看日历
    User->>FE: 点击"打卡日历"
    FE->>BE: GET /api/checkin/calendar?year=2026&month=7
    BE->>DB: 查询当月打卡记录
    DB-->>BE: 返回打卡日期列表
    BE-->>FE: 返回 JSON [[date, count], ...]
    FE-->>User: ECharts 渲染日历热力图
```

3.4 Class Diagram —

```
classDiagram
    class User {
        -Long id
        -String username
        -String password
        -String nickname
        -String avatar
        -DateTime createdAt
        +register()
        +login()
        +updateProfile()
    }

    class Task {
        -Long id
        -Long userId
        -String title
        -String description
        -Priority priority
        -DateTime deadline
        -Boolean needReview
        -Integer reviewCycle
    }
```



```

        -TaskStatus status
        -DateTime createdAt
        -DateTime updatedAt
        +create()
        +update()
        +delete()
        +getById()
        +listByUser()
    }

class CheckIn {
    -Long id
    -Long taskId
    -Long userId
    -DateTime checkTime
    -String note
    +doCheckIn()
    +getByUser()
    +getCalendar()
}

class Reminder {
    -Long id
    -Long userId
    -Long taskId
    -String message
    -DateTime remindTime
    -Boolean isRead
    +generate()
    +listByUser()
    +markRead()
}

class Recommendation {
    -Long id
    -String title
    -String description
    -Priority priority
    -String category
    +listAll()
}

class Priority {
    <<enumeration>>
    HIGH
    MEDIUM
    LOW
}

class TaskStatus {
    <<enumeration>>
    TODO

```

```

    IN_PROGRESS
    DONE
}

```

```

User "1" --> "*" Task : 创建
User "1" --> "*" CheckIn : 打卡
User "1" --> "*" Reminder : 接收
Task "1" --> "*" CheckIn : 被打卡
Task "1" --> "*" Reminder : 产生
Task ..> Priority : 使用
Task ..> TaskStatus : 使用

```

	A → B			
User → Task		1	1 : *	
Task → CheckIn		1	1 : *	
User → CheckIn		1	1 : *	
User → Reminder		1	1 : *	
Task → Reminder		1	1 : *	
Task → Priority		—	—	Task Priority
Task → TaskStatus		—	—	Task TaskStatus

4.

4.1 ER —

```

erDiagram
    User ||--o{ Task : "创建 (1:N)"
    User ||--o{ CheckIn : "打卡 (1:N)"
    User ||--o{ Reminder : "接收 (1:N)"
    Task ||--o{ CheckIn : "被打卡 (1:N)"
    Task ||--o{ Reminder : "产生 (1:N)"

    User {
        bigint id PK "用户唯一标识"
        varchar username UK "登录用户名"
        varchar password "BCrypt加密密码"
        varchar nickname "用户昵称"
        varchar avatar "头像URL"
        datetime created_at "账号创建时间"
    }

    Task {
        bigint id PK "任务唯一标识"
    }

```

```

    bigint user_id FK "所属用户ID"
    varchar title "任务标题"
    text description "任务描述"
    enum priority "HIGH/MEDIUM/LOW"
    datetime deadline "截止时间"
    tinyint need_review "是否需要复习提醒"
    int review_cycle "复习周期天数"
    enum status "TODO/IN_PROGRESS/DONE"
    datetime created_at "任务创建时间"
    datetime updated_at "最后更新时间"
}

CheckIn {
    bigint id PK "打卡记录唯一标识"
    bigint user_id FK "打卡用户ID"
    bigint task_id FK "关联任务ID"
    datetime check_time "打卡时间"
    varchar note "打卡备注"
}

Reminder {
    bigint id PK "提醒唯一标识"
    bigint user_id FK "接收提醒的用户ID"
    bigint task_id FK "关联任务ID"
    varchar message "提醒内容"
    datetime remind_time "提醒触发时间"
    tinyint is_read "是否已读 0/1"
    datetime created_at "记录创建时间"
}

Recommendation {
    bigint id PK "模板唯一标识"
    varchar title "推荐任务主题"
    text description "推荐任务描述"
    enum priority "推荐优先级"
    varchar category "分类标签"
}

```

4.2

1 user

id	BIGINT	—	N	—	PK, AUTO_INCREMENT	
username	VARCHAR	50	N	—	UNIQUE, NOT NULL	

password	VARCHAR	255	N	—	NOT NULL	BCryp
nickname	VARCHAR	50	Y	NULL	—	/
avatar	VARCHAR	255	Y	NULL	—	URL
created_at	DATETIME	—	N	CURRENT_TIMESTAMP	—	
updated_at	DATETIME	—	Y	NULL	ON UPDATE CURRENT_TIMESTAMP	

2 task

id	BIGINT	—	N	—	PK, AUT
user_id	BIGINT	—	N	—	FK
title	VARCHAR	200	N	—	NO
description	TEXT	—	Y	NULL	—
priority	ENUM('HIGH','MEDIUM','LOW')	—	N	'MEDIUM'	NO
deadline	DATETIME	—	N	—	NO
need_review	TINYINT(1)	—	N	0	—
review_cycle	INT	—	Y	NULL	—
status	ENUM('TODO','IN_PROGRESS','DONE')	—	N	'TODO'	NO
created_at	DATETIME	—	N	CURRENT_TIMESTAMP	—
updated_at	DATETIME	—	Y	NULL	ON CUF

- `idx_task_user_id` ON (user_id)
- `idx_task_status` ON (status)

- `idx_task_deadline` ON (deadline)
- `idx_task_user_status` ON (user_id, status)

3 check_in

id	BIGINT	—	N	—	PK, AUTO_INCREMENT	
user_id	BIGINT	—	N	—	FK → user.id	ID
task_id	BIGINT	—	N	—	FK → task.id	ID
check_time	DATETIME	—	N	CURRENT_TIMESTAMP	NOT NULL	
note	VARCHAR	500	Y	NULL	—	

- `idx_checkin_user_id` ON (user_id)
- `idx_checkin_task_id` ON (task_id)
- `idx_checkin_user_time` ON (user_id, check_time)
- `idx_checkin_date` ON ((CAST(check_time AS DATE)))

4 reminder

id	BIGINT	—	N	—	PK, AUTO_INCREMENT	
user_id	BIGINT	—	N	—	FK → user.id	ID
task_id	BIGINT	—	N	—	FK → task.id	ID

message	VARCHAR	500	N	—	NOT NULL	
remind_time	DATETIME	—	N	—	NOT NULL	
is_read	TINYINT(1)	—	N	0	—	(0/1)
created_at	DATETIME	—	N	CURRENT_TIMESTAMP	—	

- `idx_reminder_user_id` ON (user_id)
- `idx_reminder_user_read` ON (user_id, is_read)

5 recommendation

id	BIGINT	—	N	—	PK, AUTO_INCREMENT	
title	VARCHAR	200	N	—	NOT NULL	
description	TEXT	—	Y	NULL	—	
priority	ENUM('HIGH','MEDIUM','LOW')	—	N	'MEDIUM'	—	

category	VARCHAR	50	Y	' '	—	
----------	---------	----	---	-----	---	--

title	description	priority	category
	30	MEDIUM	
	1 LeetCode	HIGH	
	30	MEDIUM	
	30	LOW	
		HIGH	
	200	MEDIUM	

--

5.

5.1

	Windows 10+ / macOS 12+ / Linux (Ubuntu 20.04+)
JDK	JDK 17
	Maven 3.8+
	Spring Boot 3.x
ORM	MyBatis / MyBatis-Plus
	Vue 3 + Vite + Element Plus
	ECharts 5.x
HTTP	Axios
IDE	IntelliJ IDEA () + VS Code ()
	Git + GitHub / GitLab
	Postman / Apifox

5.2

OS	Ubuntu 20.04 / Windows 10	Ubuntu 22.04

	2GB RAM	4GB RAM
	1GB	5GB SSD
	MySQL 8.0+ SQLite 3	MySQL 8.0+
	Chrome 90+ / Edge 90+	Chrome
	1366×768	1920×1080

5.3

Maven pom.xml

spring-boot-starter-web	3.x	Web
spring-boot-starter-validation	3.x	
mybatis-spring-boot-starter	3.x	MyBatis
mysql-connector-j	8.x	MySQL
sqlite-jdbc	3.x	SQLite
jjwt	0.12.x	JWT
lombok		POJO
spring-boot-starter-test	3.x	

package.json

vue	3.x	
vite	5.x	
vue-router	4.x	
element-plus	2.x	UI
echarts	5.x	
axios	1.x	HTTP
pinia	2.x	

5.4

	NF-Perf-01	2
	NF-Perf-02	API 500ms (P95)

	NF-Reliability-01	
	NF-Availability-01	7×24
	NF-Usability-01	3
	NF-Security-01	BCrypt
	NF-Security-02	JWT
	NF-Security-03	6
	NF-Maintainability-01	
	NF-Extensibility-01	RESTful

5.5

	NF-Constraint-01	Java + Spring Boot
	NF-Constraint-02	Vue
	NF-Constraint-03	Git
	NF-Constraint-04	
	NF-Constraint-05	2

6.

AT-01		UC-01		
AT-02		UC-02		
AT-03		UC-03		
AT-04		UC-04		
AT-05		UC-05		
AT-06	/	UC-04		
AT-07		UC-04	/	
AT-08		UC-06	" "	
AT-09		UC-07		
AT-10		UC-07		
AT-11		UC-08	/	

AT-12		UC-08	/	
AT-13		UC-09		
AT-14		UC-09		
AT-15	JWT	—	API Token	
AT-16		—	BCrypt	
AT-17		—	A B	
AT-18		—	2	
AT-19		—	Chrome Edge	

		"ch5"				User Story	Mermaid
Flowchart	UML	/	/	Mermaid	Sequence/Class	Mermaid Flowchart	ER
		Mermaid ER				Mermaid	VS Code
		"Markdown Preview Mermaid Support"		GitHub	GitLab	Typora	